BENJAMIN MASTERS

Hearing Lab
Department of Systems Design Engineering
University of Waterloo

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EDUCATION

University of Waterloo — Waterloo, Ontario

Ph.D., Biomedical Engineering — Expected 2028

Thesis topic: Multimodal markers of cognitive effort in interactive conversation

M.A.Sc, Systems Design Engineering — 2024

Thesis: Talker sensitivity to turn-taking in conversation

Certificate in University Teaching — Expected 2026

Purdue University — West Lafayette, Indiana

B.S.E., Acoustical Engineering — 2022

RESEARCH APPOINTMENTS

University of Waterloo — Waterloo, Ontario

Graduate Researcher, Hearing Lab — 2022 – Present

- Research Area 1: Behavioral adaptations to communication difficulty
  - Designed experiments that manipulated turn-taking dynamics and speech acoustics in real-time to develop behavioral markers of listening effort and study adaptive auditory processes.
- Research Area 2: Physiological markers of cognitive effort
  - Developed approaches for utilizing eye-tracking and pupillometry as physiological indices to study the time-course of attention and effort during conversation.
- Research Area 3: Signal processing systems design
  - Developed a modular real-time signal processing framework in Python enabling the first studies to actively manipulate acoustic communication channels during conversational interactions.

Purdue University — West Lafayette, Indiana

Undergraduate Researcher, Experimental Amplification Research Lab — 2021 – 2022

- Assessed and improved the acoustic directional performance of an open-source hearing aid platform.

TEACHING APPOINTMENTS

University of Waterloo — Waterloo, Ontario

Course Instructor

- Linear Systems & Signals — Spring 2025 & 2026
  - Served as the sole instructor for a core undergraduate engineering course with ~80 students.
  - Responsibilities included the design and delivery of all lecture materials and assessments and the managing of two teaching assistants.
  - Achieved a teaching evaluation of 4.7/5.

Teaching Assistant

- Biomedical Signal Processing — Winter 2026
- Linear Systems & Signals — Several Terms 2023 – 2025
- Computational Neuroscience — Winter 2025
- Simulating Neurobiological Systems — Fall 2023
- Communication in Biomedical Engineering — Fall 2022

Purdue University — West Lafayette, Indiana

Teaching Assistant

- Noise Control — Spring 2022

INDUSTRY EXPERIENCE

Bose Corporation Framingham, Massachusetts
Acoustic Engineer Intern 2022
- Developed an automated algorithm and testing procedure to acoustically detect air leaks from loudspeaker enclosures.

Starkey Hearing Technologies Eden Prairie, Minnesota
Acoustic Engineer Intern 2021
- Adapted vibration testing procedures for electroacoustic transducers to improve coherence of measurements.

AWARDS

- Acoustical Society of America Student Travel Award 2025
- International Symposium on Auditory and Audiological Research Scholarship 2025
- 1st Place - International Student Challenge in Signal Processing - Acoustical Society of America 2023 & 2024
- International Masters Award of Excellence - University of Waterloo 2022 – 2024
- Canadian Acoustical Association Student Travel Award 2024

SERVICE

- Session Co-chair - Undergraduate and Graduate Programs in Acoustics - Acoustical Society of America 2026
- Student Council Representative - Signal Processing Committee - Acoustical Society of America 2025 – 2027
- Organizing Committee - Symposium on Aging Research 2025 – 2026
- Biomedical Engineering Program Representative - Various Graduate Studies Fairs 2025
- Committee Member - Systems Design/Biomedical Engineering Teaching Assistant Spaces 2025
- Treasurer - Purdue Chapters of the Acoustical Society of America and Audio Engineering Society 2020 – 2022
- Vice-President and Treasurer - Purdue University Multidisciplinary Engineering Student Society 2019 – 2021

SCHOLARSHIP

Journal Articles

[J3] Hearing loss is reflected in conversational pupil responses
Masters, B., Shiell, M.M., Wendt, D., MacDonald, E.N.
Under Review

[J2] Variable delay affects conversational turn-taking behavior in the presence of background noise
Masters, B., MacDonald, E.N.
The Journal of the Acoustical Society of America, 2025

[J1] Pupil responses during interactive conversation
Masters, B., Aliakbaryhosseinabadi, S., Wendt, D., MacDonald, E.N.
Trends in Hearing, 2025

Conference Presentations & Posters

[C10] The impacts of asymmetric acoustic conditions on speech production and communication dynamics during interactive conversation
Masters, B., Stepanenko P., MacDonald, E.N.
Acoustical Society of America 190, Philadelphia, Pennsylvania, 2026

[C9] Pupillometry reveals differences in the cognitive demands of interactive conversation versus passive listening
Masters, B., Shiell, M., Wendt, D., MacDonald, E.N.
Acoustical Society of America 189, Honolulu, Hawaii, 2025

[C8] Assessing the sensitivity of the pupil dilation response to different sources of difficulty in interactive conversation
Masters, B., Stepanenko, P., Shiell, M., Wendt, D., MacDonald, E.N.
Acoustical Society of America 189, Honolulu, Hawaii, 2025

[C7] Changes in conversational behavior across noise, language proficiency, and task
Stepanenko, P., **Masters, B.**, MacDonald, E.N.
Acoustical Society of America 189, Honolulu, Hawaii, 2025

[C6] Preliminary investigation of the effect of noise on semantic content of conversations
Stepanenko, P., **Masters, B.**, MacDonald, E.N.
Acoustical Society of America 189, Honolulu, Hawaii, 2025

[C5] The effects of hearing loss on pupil responses in conversation
Masters, B., Shiell, M., Wendt, D., MacDonald, E.N.
International Symposium on Auditory and Audiological Research, Nyborg, Denmark, 2025
Symposium on Aging Research, Waterloo, Ontario, 2025 ***(Research Institute for Aging Award)***

[C4] Understanding effortful listening in interactive conversation with pupillometry
Masters, B., Shiell, M., Wendt, D., MacDonald, E.N.
Workshop in Communication in Hearing Science, Lynge, Denmark, 2025
Biomedical Engineering Research Day, Waterloo, Ontario, 2025 ***(Best Presentation Award)***

[C3] Analysis of physiological measures around conversational state changes
Masters, B., Aliakbaryhosseinabadi, S., Wendt, D., MacDonald, E.N.
Acoustical Society of America 186/Acoustics Week in Canada, Ottawa, Ontario, 2024

[C2] The effects of delay on task-based interactive conversation
Masters, B., MacDonald, E.N.
Acoustical Society of America 186/Acoustics Week in Canada, Ottawa, Ontario, 2024

[C1] Directionality characteristics of the Tympan open-source hearing aid and earpieces
Masters, B., Alexander, J.M.
Acoustical Society of America 181, Seattle, Washington, 2021

Seminars & Invited Talks

[S2] Inferring effort in conversation: Behavioral and physiological markers
Sonova, 2026

[S1] Talker Sensitivity to Turn-Taking in Conversation
Eriksholm Research Centre, 2024
Technical University of Denmark, 2024

Open Datasets

[D2] Replication Data for Variable delay affects conversational turn-taking in the presence of background noise
Masters, B., MacDonald, E.N.
Borealis, 2025

[D1] Replication Data for Pupil responses during interactive conversation
Masters, B., Aliakbaryhosseinabadi, S., Wendt, D., MacDonald, E.N.
Borealis, 2025

Theses

[T1] Talker Sensitivity to Turn-Taking in Conversation
M.A.Sc. Systems Design Engineering, University of Waterloo, 2024